TRIASHA SARKAR

Atlanta, GA | +1 (404) 948-8761 | tsarkar34@gatech.edu | linkedin.com/in/triasha-sarkar | github.com/triasha72 | triasha72.github.io/Portfolio

Machine Learning Engineer with experience in scientific ML, model evaluation, retrieval, time-series data, and ML systems. Aerospace background in simulation, uncertainty, and engineering problems.

EXPERIENCE

Graduate Research Assistant | Georgia Tech ASDL | May 2025 – Aug 2026

- Built ML workflows for simulation reliability, airline-safety retrieval, and sustainable aviation; documented assumptions and results for faculty and sponsors.

Machine Learning Engineer | Rolls-Royce | Jul 2023 – Apr 2025

- Built Python workflows for engine diagnostics, anomaly detection, predictive maintenance, and mixed-frequency sensor data; reviewed findings with lifecycle engineers.

Data Science Intern | Rolls-Royce DataLabs | May 2021 – Jul 2021

- Cleaned aircraft-engine sensor data, built features, and compared prediction and anomaly-detection models for engineering review.

PROJECTS

AIRFAANS | CFD surrogates

- Compared an MLP, MeshGraphNet-style GNN, and point neural operator on 200 AirfRANS meshes per model across three seeds; kept OOD and uncertainty work explicitly pending.

EdgeGenBench | Flight-anomaly ML and ONNX

- Evaluated NASA DASHlink data on 17,780 held-out aircraft approaches; 0.738 macro F1 missed the release target, while ONNX consistency stayed above 99.55% under tested sensor corruptions.

NewsLens | Recommendation and real-time search

- Built a leakage-aware news recommender and Go, Kafka, PostgreSQL, and FastAPI ingestion path; NDCG@10 0.3664 and sampled index-freshness p95 79 ms.

Surrogate Model Learning | Reliability under design shift

- Used grouped splits and uncertainty checks on public engineering data; airfoil GP reached R^2 0.8145, while nominal 90% intervals lost coverage under design shift.

Other ML projects | IntegrityBench | AeroRAG-X | AeroSynth-Eval | Equity Backtest

- Built evaluation and release controls for changing moderation policies; source-grounded NASA technical-report retrieval; synthetic inspection-image evaluation; and a walk-forward equity-signal backtest.

Georgia Tech ASDL projects | Rocket-Motor Failure Detection | GREEN TEA and Project EAGLE | HERO

- Worked on silent-failure detection in simulation, sustainable-aviation modeling, and source-aware retrieval for airline safety.

EARLIER AEROSPACE PROJECTS

Supersonic Business Jets and Noise Mitigation (Cranfield) | Bird-Strike Impact Analysis (SRM) | Ornithopter Aerodynamics (SRM)

- Studied geometry and noise trade-offs for supersonic aircraft; used ANSYS to compare nose-cone materials for bird-strike impact; used CFD to examine ornithopter wing area and load factor.

EDUCATION & SKILLS

MS Aerospace Engineering, Georgia Tech (2026) | MSc Aerospace Vehicle Design, Cranfield (2023) | B.Tech Aerospace Engineering, SRM (2021)

Python, PyTorch, PyTorch Geometric, scikit-learn, GNNs, surrogate modeling, uncertainty quantification, SQL, FastAPI, Docker, Kubernetes, Kafka, PostgreSQL, ONNX

LEADERSHIP

Graduate Chair, Women of Aeronautics and Astronautics, Georgia Tech | Aaruush organizing team (Jul 2019 – Sep 2019) and SRM hackathon participant (Sep 2019)